

SMART INDIA HACKATHON 2026



Problem Statement ID - SIH26094

Problem Statement Title - AI-Powered Dynamic Mental Health

Monitoring and Distress Prediction System for Victims of Atrocities

Theme - MedTech / BioTech / HealthTech

PS Category - Software

Team ID - <fill in your registered Team ID>

Team Name - JAMMERZ



AI-Powered Mental Health Monitoring for Atrocity Victims

PROPOSED SOLUTION

Proposed Solution

AI system that **continuously monitors** victims registered via **NHAA 14566 / Integrated Portal**

Regular well-being checks - chatbot, IVRS, SMS, mobile app

AI analysis - NLP, sentiment, emotion, voice stress, behaviour/engagement

Dynamic Distress Score + trend + prediction - is distress increasing?

Real-time alerts to counsellors/officials when risk is high

Recommended interventions - counselling, medical, financial aid

District / State / National dashboards, explainable AI, privacy & security

5 Core Modules

Victim Module - login, dashboard, well-being check-in

Mental Health Monitoring Module - questionnaire, multilingual chatbot/IVRS/SMS

AI Risk Engine - sentiment/voice-stress analysis, distress score, prediction

Official/Counsellor Module - risk dashboard, case details, intervention tracking

Administration & Security Module - district/state/national dashboards, RBAC

Tech Stack

Client: React Native - one codebase, Mobile + Web

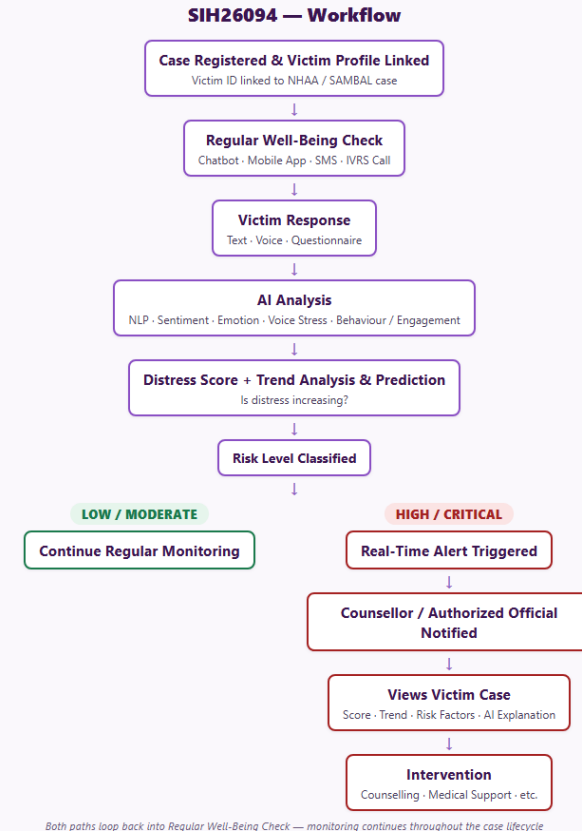
Backend: Node.js + Express

Platform: Supabase - Postgres, Auth, Storage, Realtime

AI: Gemini (chat/sentiment) + Google Speech/Bhashini (ASR/TTS) + SpeechBrain (voice stress)

Backend AI Features

AI response analysis → **Distress Score Engine** → history/trend → **prediction** → risk classification → **automated case prioritisation**



Feasibility

Mature, well-documented stack - React Native, Node, Supabase

Pretrained SER models available today - no training needed

Extends NHAA/SAMBAL + Tele-MANAS - not built from scratch

One codebase → **feature parity by design**

Challenges

No real victim data available (ethically/legally)

Voice-stress models trained on **English** - India-language gap

Highly sensitive data - **privacy/security risk**

False positive/negative risk in crisis prediction

Strategies to Overcome

Validate on **public datasets + synthetic personas**

Human-in-the-loop - AI recommends, never auto-acts

DPDP-compliant architecture from day one

IMPACT AND BENEFITS

Impact

Closes the crisis-detection gap - flags distress weeks earlier
Strengthens victim trust in the justice system
Gives policymakers **real visibility at scale**
Improves welfare / counselling / law-enforcement **coordination**

Benefits

Victims: multilingual, voice-first - reaches rural/semi-literate users
Counsellors: automated prioritisation + AI case-note copilot
MoSJE: reuses existing infra - faster, cheaper rollout
Protects highly sensitive data by design

Government Systems Referenced

PIB - 14566: National Helpline Against Atrocities - pib.gov.in
NHAA / SAMBAL portal - nhapoa.dosje.gov.in
Tele-MANAS official portal, MoHFW - telemanas.mohfw.gov.in
SC/ST (Prevention of Atrocities) Act, 1989 - ncsk.nic.in
Witness Protection Scheme, 2018 - IJLMH

Research and Case Studies

India's Tele-MANAS: evolution, early outcomes - **BJPsych / Cambridge Core**
Tele-MANAS callers turning to humans after AI falls short - **Vision IAS / The Hindu**
Speech Emotion Recognition in Mental Health: Systematic Review - **JMIR, 2025**
AI-Powered Holistic Mental Health Monitoring - **ResearchGate, 2025**
Crisis Text Line - Detecting Crisis: An AI Solution - **~85-90%** severe-risk detection from text alone